

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]

Tap amplitudes (pixels): [146.40013661195812, 127.00393694685216, 117.00427342623003, 127.7810627597063, 131.36590120727678, 141.65097952361643, 117.92370414806346, 122.44182292011173, 120.0, 126.21014222319853, 118.92854997854805]



Analysis of Extracted Features

To determine the MDS-UPDRS Item 3.4 score, we must evaluate the data against the three primary pathological features: **Speed (Slowing)**, **Rhythm (Interruptions/Hesitations)**, and **Amplitude (Decrementing/Sequence Effect)**.

1. Speed & Rhythm (Tap Periods)

- **Data:** [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]
- **Observation:** The periods fluctuate (range 141–207 frames), but there is **no progressive slowing trend** (the periods do not get consistently longer throughout the test). The final period (170) is actually faster than the first (189).
- **Hesitations:** There are no extreme outliers (e.g., a period of 400+) that would indicate a "halt" or "freeze." The variance observed (alternating between ~150 and ~200) suggests some rhythm irregularity, but it does not constitute distinct interruptions or arrests.

2. Amplitude (Decrements)

- **Data:** [146.4, 127.0, 117.0, 127.8, 131.4, 141.7, 117.9, 122.4, 120.0, 126.2, 118.9]
- **Observation:** The amplitude starts high (146.4), dips initially, but **recovers significantly** midway (Tap 6 reaches 141.7, nearly matching the start).
- **Decrement Criteria:**
 - *Score 1 (Near end):* The last 5 taps are stable (hovering around 118–126), not fading out.
 - *Score 2 (Midway):* The amplitude actually increases/recovers midway rather than decrementing.
 - *Score 3 (After 1st tap):* While there is a drop after the 1st tap, the subsequent recovery at Tap 4, 5, and 6 rules out the pathological "sequence effect" (continuous, unrecoverable fatigue) typical of Parkinsonism.

3. Tap Count

- **Data:** 11 Taps
- **Observation:** The subject successfully completed the requested 10 taps.

Scoring Rationale

The subject demonstrates the ability to maintain speed without fatigue. While the amplitude settles slightly lower than the very first tap, the midway recovery demonstrates that this is not a pathological decrement (sequence effect). The rhythm variances are within a functional range and do not meet the criteria for distinct interruptions or halts.

Score: 0